LECTURE 11:
MARTINGALES - 1

The goal of these lecture notes is to give an exposition of the material in the same order and approximately in the same completeness as it was done in class (ORF526). Always refer to the textbooks if in doubt and for more complete treatment.

Please email me with any observed typos to `elre@princeton.edu`, thanks in advance!

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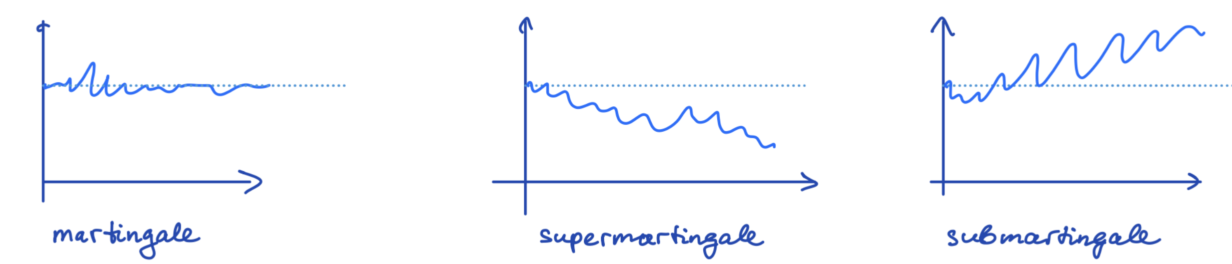
Definition 11.1 (**Martingale process**). Let $(X_n)_{n\geq 0} = (X_0, X_1, X_2, \ldots)$ be a discrete time stochastic process, and $(\mathcal{F}_n)_{n\geq 0}$ be a **filtration**, that is, a sequence $\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \ldots$ of sub-σ-algebras. We say that $(X_n)_{n\geq 0}$ is a **martingale** with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$ if

(i) $(X_n)_{n\geq 0}$ is **adapted** to $(\mathcal{F}_n)_{n\geq 0}$: that is, X_n is $\mathcal{F}_n$-measurable for all n.
(ii) X_n is integrable for all n.
(iii) $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = X_n$ for all n.

The **natural filtration** associated with a stochastic process $(X_n)_{n\geq 0}$ is given by $\mathcal{F}_n := \sigma(X_0, X_1, \ldots, X_n)$ for all n. If the filtration for a martingale is not specified, it is assumed that the natural filtration is used.

Finally, we say that a stochastic process $(X_n)_{n\geq 0}$ is

- a **supermartingale** if it satisfies properties (i), (ii) and (iii'): $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \leq X_n$ for all n.
- a **submartingale** if it satisfies properties (i), (ii) and (iii"): $\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] \geq X_n$ for all n.
- **predictable** if X_n is $\mathcal{F}_{n-1}$ measurable for all n.



Basic properties

Lemma 11.2. *Let $(X_n)_{n\geq 0}$ be a martingale. Then for all $n, m \geq 0$, $\mathbb{E}[X_{n+m} \mid \mathcal{F}_n] = X_n$.*

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Proof. Fix $n \geq 0$. We will show this inductively on m. The base case $m = 1$ follows from the martingale property (iii). Assuming that the claim is true for m, we want to prove that it holds for $m + 1$. This follows from

$$\begin{aligned} \mathbb{E}[X_{n+m+1} \mid \mathcal{F}_n] &= \mathbb{E}[\mathbb{E}[X_{n+m+1} \mid \mathcal{F}_{n+m}] \mid \mathcal{F}_n] && \text{(tower property)} \\ &= \mathbb{E}[X_{n+m} \mid \mathcal{F}_n] && \text{(martingale property)} \\ &= X_n && \text{(inductive step).} \end{aligned}$$

□

Lemma 11.3. *Let $(X_n)_{n\geq 0}$ be a martingale. Then for all $n, m \geq 0$, $\mathbb{E}[X_{n+m}] = \mathbb{E}[X_n]$.*

Proof. Take the expectation in Lemma 11.2 and use the law of total expectations (i.e., tower property). □

Lemma 11.4 (Doob decomposition)**.** *Let $(X_n)_{n\geq 0}$ be an adapted process with respect to the filtration $(\mathcal{F}_n)_{n\geq 0}$. Then there exists a martingale $(M_n)_{n\geq 0}$ and an integrable predictable process $(A_n)_{n\geq 0}$ with $M_0 = A_0 = 0$ such that*

$$X_n = X_0 + M_n + A_n \quad \textit{for all } n.$$

That is, X_n admits a decomposition into a predictable part (i.e., drift) and a martingale component (i.e., diffusion).

Proof. We can decompose $X_n - X_0$ as a telescoping sum for each n:

$$X_n - X_0 = \underbrace{\sum_{j=1}^{n}(X_j - \mathbb{E}[X_j \mid \mathcal{F}_{j-1}])}_{:=M_n} + \underbrace{\sum_{j=0}^{n-1}(\mathbb{E}[X_{j+1} \mid \mathcal{F}_j] - X_j)}_{:=A_n}.$$

By construction, A_n is $\mathcal{F}_{n-1}$ measurable and hence predictable. We can also verify that M_n is a martingale: using the tower property,

$$\begin{aligned} \mathbb{E}[M_n \mid \mathcal{F}_{n-1}] &= \sum_{j=1}^{n}(\mathbb{E}[X_j \mid \mathcal{F}_{n-1}] - \mathbb{E}[\mathbb{E}[X_j \mid \mathcal{F}_{j-1}] \mid \mathcal{F}_{n-1}]) \\ &= \sum_{j=1}^{n-1}(X_j - \mathbb{E}[X_j \mid \mathcal{F}_{j-1}]) + (\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] - \mathbb{E}[X_n \mid \mathcal{F}_{n-1}]) = M_{n-1}. \end{aligned}$$

□

In Lemma 11.4, if $(X_n)_{n\geq 0}$ is a supermartingale (resp. submartingale), then A_n is decreasing (resp. increasing). (Exercise!)

Furthermore, the decomposition is unique up to the initial value of X_0. Indeed, if $X_n = X_0 + M_n + A_n = X_0 + \widetilde{M}_n + \widetilde{A}_n$, then

$$\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}[M_n - M_{n-1} \mid \mathcal{F}_{n-1}] + \mathbb{E}[A_n - A_{n-1} \mid \mathcal{F}_{n-1}] = A_n - A_{n-1}.$$

Similarly, $\mathbb{E}[X_n - X_{n-1} \mid \mathcal{F}_{n-1}] = \widetilde{A}_n - \widetilde{A}_{n-1}$. So, $A_n - A_{n-1} = \widetilde{A}_n - \widetilde{A}_{n-1}$ a.s. for all n, which implies that $\widetilde{A}_n = A_n$ and also $\widetilde{M}_n = M_n$ for all n.

Examples

Sums of independent mean-zero random variables. Let $X_1, X_2, \dots$ be independent integrable random variables with $\mathbb{E}X_i = 0$ for all i. Then $(S_n)_{n\geq 0}$, with $S_n := \sum_{i=1}^n X_i$ and $S_0 := 0$ is a martingale with respect to $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ ($\mathcal{F}_0 = \{\emptyset, \Omega\}$). Why?

- S_n is $\mathcal{F}_n$-measurable (as $X_1, \dots, X_n$ are $\mathcal{F}_n$-measurable).
- S_n is integrable (finite sum of integrable X_i's.
- $\mathbb{E}[S_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[X_n \mid \mathcal{F}_{n-1}] + \mathbb{E}[S_{n-1} \mid \mathcal{F}_{n-1}] = \mathbb{E}X_n + S_{n-1} = S_{n-1}$ since X_n is independent of $\mathcal{F}_{n-1}$, and S_{n-1} is $\mathcal{F}_{n-1}$-measurable.

Products of independent mean-one random variables. Let $X_1, X_2, \dots$ be independent integrable random variables such that $X_i \geq 0$ a.s., and $\mathbb{E}X_i = 1$ for all i. Then $M_0 = 1$, $M_n := \prod_{i=1}^n X_i$ is a martingale with respect to $\mathcal{F}_n = \sigma(X_1, \dots, X_n)$ ($\mathcal{F}_0 = \{\emptyset, \Omega\}$). Why?

- Think through (i) and (ii)!
- $\mathbb{E}[M_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[X_n M_{n-1} \mid \mathcal{F}_{n-1}] = M_{n-1}\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = M_{n-1}\mathbb{E}[X_n] = M_{n-1}$.

Doob martingale. Let X be an integrable random variable, and $(\mathcal{F}_n)_{n\geq 1}$ be any filtration. Then $X_n := \mathbb{E}[X \mid \mathcal{F}_n]$ is a martingale. Why?

- (i) + (ii): conditional expectation with respect to $\mathcal{F}_n$ is $\mathcal{F}_n$-measurable and integrable by definition.
- $\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] = \mathbb{E}[\mathbb{E}[X \mid \mathcal{F}_n] \mid \mathcal{F}_{n-1}] = \mathbb{E}[X \mid \mathcal{F}_{n-1}] = X_{n-1}$ by the tower property for the second-to-last equality.

Martingale transform. Let $(X_n)_{n\geq 0}$ be an adapted process and $(A_n)_{n\geq 0}$ be a predictable process with respect to a filtration $(\mathcal{F}_n)_{n\geq 0}$. Define $(Y_n)_{n\geq 0}$ by $Y_0 := 0$, and $Y_n := \sum_{k=1}^n A_k(X_k - X_{k-1})$. We denote this process by $Y = A \bullet X$. When is Y a martingale?

Lemma 11.5. *Let $(A_n)_{n\geq 0}$ be a bounded predictable process: $|A_n| \leq K$ for all n for some constant K. If $(X_n)_{n\geq 0}$ is a martingale, then $Y = A \bullet X$ is also a martingale. If $A_n \geq 0$ and $(X_n)_{n\geq 0}$ is a super- or submartingale, then $Y = A \bullet X$ is also a super- or submartingale respectively.*

Proof. Suppose that X_n is a martingale. Then

$$\begin{aligned}
\mathbb{E}[Y_n - Y_{n-1} \mid \mathcal{F}_{n-1}] &= \mathbb{E}[A_n(X_n - X_{n-1}) \mid \mathcal{F}_{n-1}] \\
&= A_n(\mathbb{E}[X_n \mid \mathcal{F}_{n-1}] - X_{n-1}) \qquad && (A_n \text{ is } \mathcal{F}_{n-1}\text{-measurable}) \\
&= 0 && (X_n \text{ is a martingale}). \qquad (1)
\end{aligned}$$

If A_n is bounded, then Y_n is integrable. Y_n is $\mathcal{F}_n$-measurable since $X_1, \dots, X_n$ and $A_1, \dots, A_n$ are. The case where X_n is a super- or submartingale follows in the same way with the respective inequality at the line (1). $\square$

Remark 11.6 (Interpretation of martingale transforms)**.** We say that Y is a martingale transform of X by A, denoted $Y = A \bullet X$:

(a) "Y = integral $\int M \, \mathrm{d}X$"
(a) A_n = your stake at game n, X_n = random outcome of game n, and Y_n = your total winning after game n.

In general, this is a common and important interpretation of martingale processes, as the rewards in consecutive games or the profits obtained from the investments that change the values with time. We will be returning to it through several examples below.

Limiting behavior

Question: What happens with a martingale process in the long run?

Theorem 11.7 (Martingale convergence theorem). *Suppose that $(X_n)_{n\geq 0}$ in $(\Omega, \mathcal{F}, \mathbb{P})$ is a supermartingale such that $\mathbb{E}|X_n| \leq K$ for all n for some constant K. Then there exists a random variable $X_\infty \in L^1(\Omega, \mathcal{F}, \mathbb{P})$ such that $X_n \to X_\infty$ pointwise as $n \to \infty$ a.s.*

So, this martingale convergence theorem shows when the *almost sure* limit of a martingale sequence must exist and be integrable (in particular, the limit cannot be infinite, except from possibly a set of measure zero). A sufficient condition is uniform bound on expectations. Note that every X_n must have finite expectation by definition of a martingale, but it does not always imply that all X_n have the same ("uniform") finite upper bound on their expectations.

Other convergence results for martingales can be found in, e.g., [Dembo, Section 5.3] and [Chatterjee, Section 9.13]. For example, uniform integrability implies convergence in L_1 for martingales – compare this with Theorem 7.4 from Lecture 7.

First, we will prove a technical tool that we will be needed (see an illustration on the next page):

Lemma 11.8 (Doob's upcrossing inequality). *Let $(X_n)_{n\geq 0}$ be a supermartingale. For real numbers $a < b$, define $U_n(a,b)$ to be the number of* upcrossings *of the interval $[a,b]$ in the time interval $[0,N]$:*

$$U_N(a,b) := \max\{k \mid \exists s_1, t_1, \ldots, s_k, t_k : 0 \leq s_1 < t_1 < s_2 < \ldots < t_k \leq N,\ X_{s_i} < a,\ X_{t_i} > b\ \forall i\}.$$

Then

$$(b-a) \cdot \mathbb{E}[U_N(a,b)] \leq \mathbb{E}[(X_n - a)_-].$$

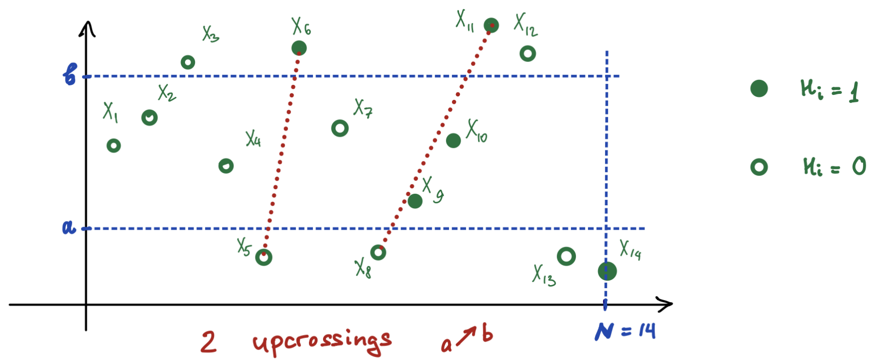
Proof. Define $H_1 := \mathbb{1}_{\{X_0 < a\}}$, and $H_{n+1} := \mathbb{1}_{\{H_n = 1\}} \cdot \mathbb{1}_{\{X_n \leq b\}} + \mathbb{1}_{\{H_n = 0\}} \cdot \mathbb{1}_{\{X_n < a\}}$ for $n \geq 1$. Note that H_n is a predictable process. This can be interpreted as the process in which you bet one dollar if you observe X_n to fall below a, and continue to bet one dollar for each time unit afterwards until X_n crosses above b.

Let $Y = H \bullet X$ be the martingale transform. Since X_n is a supermartingale and H_n is bounded by one, $\{Y_n\}$ is also a supermartingale by Lemma 11.5.

Therefore, $\mathbb{E}Y_N \leq \mathbb{E}Y_0 = 0$. Furthermore, we have

$$Y_N \geq (b-a) \cdot U_N(a,b) - (X_N - a)_-.$$

This is because every upcrossing increases Y by at least $(b-a)$, and $(X_n - a)_-$ is an adjustment for a possible negative amount that is subtracted at the end. Taking expectations and rearranging completes the proof. □



$H_6, H_9, H_{10}, H_{11}, H_{14} = 1,$ others $= 0$

$$Y_N = \sum_{i=1}^{14} H_i (X_i - X_{i-1}) = (X_6 - X_5) + (X_{11} - X_8) + (X_{14} - X_{13}) \geqslant 2 \cdot (b-a) + \underbrace{(X_{14} - a)}_{\leq 0}$$

We can now prove the pointwise martingale convergence theorem.

Proof of Theorem 11.7. Let $\mathcal{E}$ be the event that X_n does not converge to a pointwise limit (potentially $\pm\infty$):

$$\begin{aligned} \mathcal{E} &= \{\omega \in \Omega : \liminf_n X_n(\omega) \neq \limsup_n X_n(\omega)\} \\ &= \bigcup_{\substack{a<b \\ a,b\in\mathbb{Q}}} \{\omega \in \Omega : \liminf_n X_n(\omega) < a < b < \limsup_n X_n(\omega)\}. \end{aligned}$$

For any given pair a, b, observe that by definition of the limsup and liminf (i.e., there are infinite sequences converging to each of them), we have the following connection to the number of upcrossings:

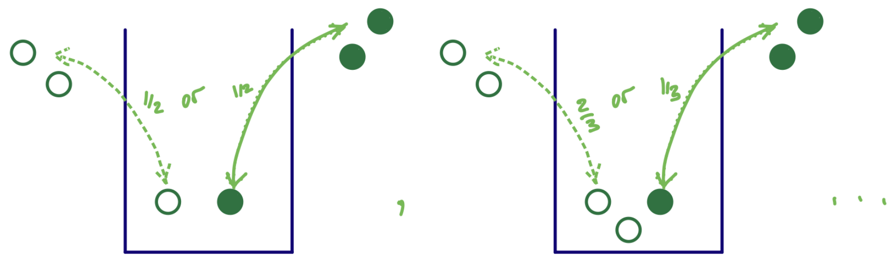
$$\{\liminf_n X_n < a < b < \limsup_n X_n\} \subseteq \{U_\infty(a,b) = \infty\} = \lim_{n\to\infty} U_n(a,b).$$

However, by Doob's upcrossing inequality (Lemma 11.8),

$$(b-a) \cdot \mathbb{E}U_n(a,b) \leq \mathbb{E}(X_n - a)_- \leq \mathbb{E}|X_n| + |a| \leq K + a < \infty.$$

Therefore, by the MCT, $\mathbb{E}U_\infty(a,b) < \infty$, so $U_\infty(a,b) < \infty$ a.s., which shows that $\mathbb{P}(\mathcal{E}) = 0$ and hence X_n converges to a pointwise limit X_∞ a.s. Finally, by Fatou's lemma, $\mathbb{E}|X_\infty| = \mathbb{E}\liminf_n |X_n| \leq \liminf_n \mathbb{E}|X_n| \leq K$, so X_∞ is integrable and finite a.s. $\square$

Example 11.9 (**Pólya's urn**). *Suppose that we have an urn with one white ball and one black ball. At each time step, we pick a ball from the urn uniformly at random and replace the ball with two balls of the same color. What is the distribution of the color of the balls in the urn after repeating this many times?*



Let's find a martingale here. More precisely, for $n \geq 2$, let X_n denote the number of white balls in the urn when there are n balls in the urn (i.e., after $n-2$ draws), so that $X_2 = 1$, and let $\mathcal{F}_n$ be the natural filtration with respect to X_n. Note that

$$\mathbb{E}[X_{n+1} \mid \mathcal{F}_n] = (X_n + 1)\frac{X_n}{n} + X_n\left(1 - \frac{X_n}{n}\right) = X_n\left(1 + \frac{1}{n}\right),$$

which shows that X_n is a submartingale. Let W_n denote the fraction of white balls in the urn when there are n balls in the urn, so that $W_2 = 1/2$. Then

$$\begin{aligned}\mathbb{E}[W_{n+1} \mid \mathcal{F}_n] &= \frac{nW_n + 1}{n+1}W_n + \frac{nW_n}{n+1}(1 - W_n) \\ &= \frac{nW_n^2}{n+1} + \frac{W_n}{n+1} + \frac{nW_n}{n+1} - \frac{nW_n^2}{n+1} = W_n.\end{aligned}$$

Thus, W_n is a martingale! Clearly $\mathbb{E}|W_n| \leq 1$ for all n. What is the almost sure pointwise limit of W_n as guaranteed by Theorem 11.7?

Let's see what happens in distribution. When there are $n \geq 2$ balls in the urn (and thus $n-2$ draws), we have for $0 \leq m \leq n-1$,

$$\begin{aligned}\mathbb{P}(X_n = m) &= \frac{(1)(2)\ldots(m-1)\cdot(1)(2)\ldots(n-m-1)}{(2)(3)\ldots(n-1)}\binom{n-2}{m-1} \\ &= \frac{(m-1)!(n-m-1)!}{(n-1)!}\cdot\frac{(n-2)!}{(m-1)!(n-m-1)!} = \frac{1}{n-1}.\end{aligned}$$

This is because of the $n-2$ draws, we need to select $m-1$ white balls, and $n-m-1$ black balls. The key observation is that all sequences of draws to get to such a combination have the same probability (we say that the sequence of draws forms an *exchangeable* sequence of random variables). This shows that after $n-2$ draws, X_n is uniform on $\{2, 3, \ldots, n-1\}$, and therefore W_n is also uniform on $\{\frac{2}{n}, \frac{3}{n}, 1 - \frac{1}{n}\}$. This allows us to deduce that $W_n \to W_\infty$, where $W_\infty \sim \mathrm{Unif}(0, 1)$!